FIFO Page Replacement Algorithm

Write a C program to implement the FIFO Page Replacement Algorithm. The program should take the number of frames and a reference string (sequence of page requests) as input. For each page in the reference string, determine if it results in a 'Page Hit' or a 'Page Fault'. Display the state of the frames after each request and calculate the total number of page faults and hits.

Input Format

1. The first line of input contains an integer representing the number of pages in the reference string.
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The total number of Page Faults and Page Hits are displayed

Sample Input 1

7

1 3 0 3 5 6 3

3

Sample Output 1

Enter number of pages in reference string:

Enter the reference string:

Enter number of frames:

Total Page Faults: 6

Total Page Hits: 1

How it works

```
Ref String | Frames | Status
-----------|--------|-------
1          | 1 - -  | Fault
3          | 1 3 -  | Fault
0          | 1 3 0  | Fault
3          | 1 3 0  | Hit
5          | 5 3 0  | Fault
6          | 5 6 0  | Fault
3          | 5 6 3  | Fault
```

Solution:

```c
#include <stdio.h>

int main() {
    int pages[100], frames[10];
    int n, f, i, j;
    int page_faults = 0, page_hits = 0;
    int index = 0;

    // Input
    printf("Enter number of pages in reference string: \n");
    scanf("%d", &n);

    printf("Enter the reference string: \n");
    for(i = 0; i < n; i++) {
```

```
    scanf("%d", &pages[i]);
}

printf("Enter number of frames: \n");
scanf("%d", &f);

// Initialize frames
for(i = 0; i < f; i++) {
    frames[i] = -1;
}

// FIFO Algorithm
for(i = 0; i < n; i++) {
    int found = 0;

    // Check HIT
    for(j = 0; j < f; j++) {
        if(frames[j] == pages[i]) {
            found = 1;
            page_hits++;
            break;
        }
    }

    // If FAULT
    if(!found) {
        frames[index] = pages[i];
        index = (index + 1) % f;
```

```
        page_faults++;
      }
    }

    // Output
    printf("\nTotal Page Faults: %d\n", page_faults);
    printf("Total Page Hits: %d\n", page_hits);

    return 0;
}
```